GLD / GDX Pairs Trading

Cointegration Analysis & Kalman Filter Spread | 2010-2024

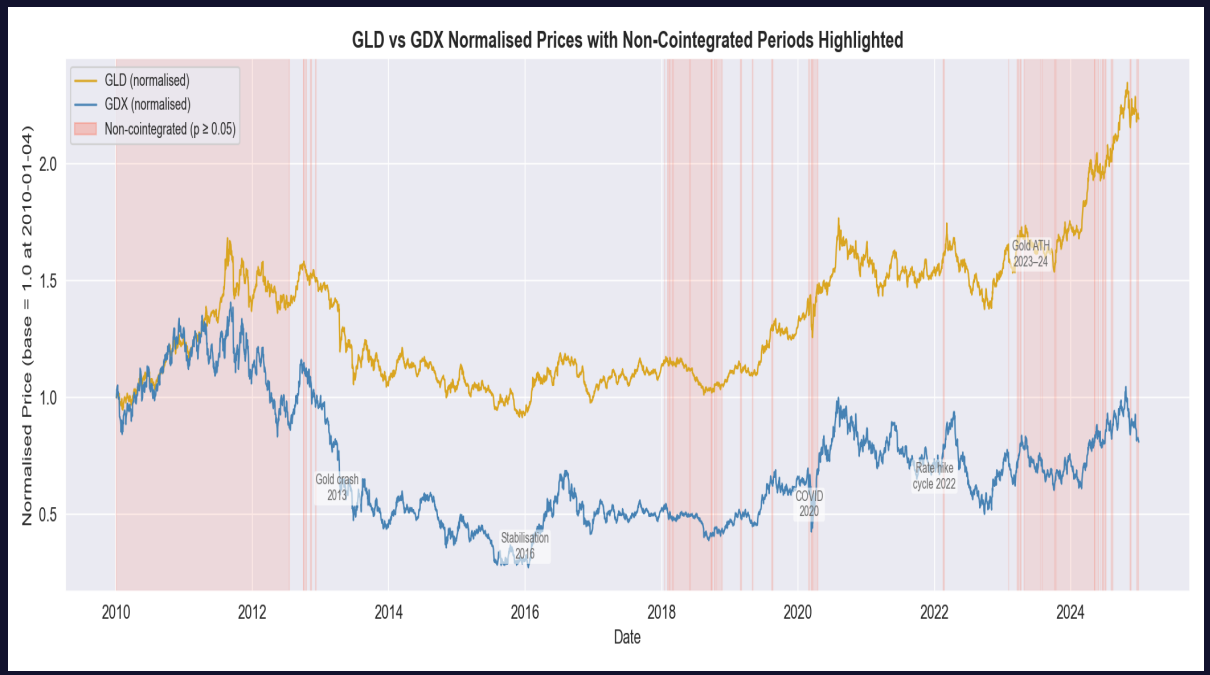

Fig 1: Normalised prices. Red = non-cointegrated windows (rolling EG p ≥ 0.05). Divergence post-2013 drives the full-period non-cointegration result.

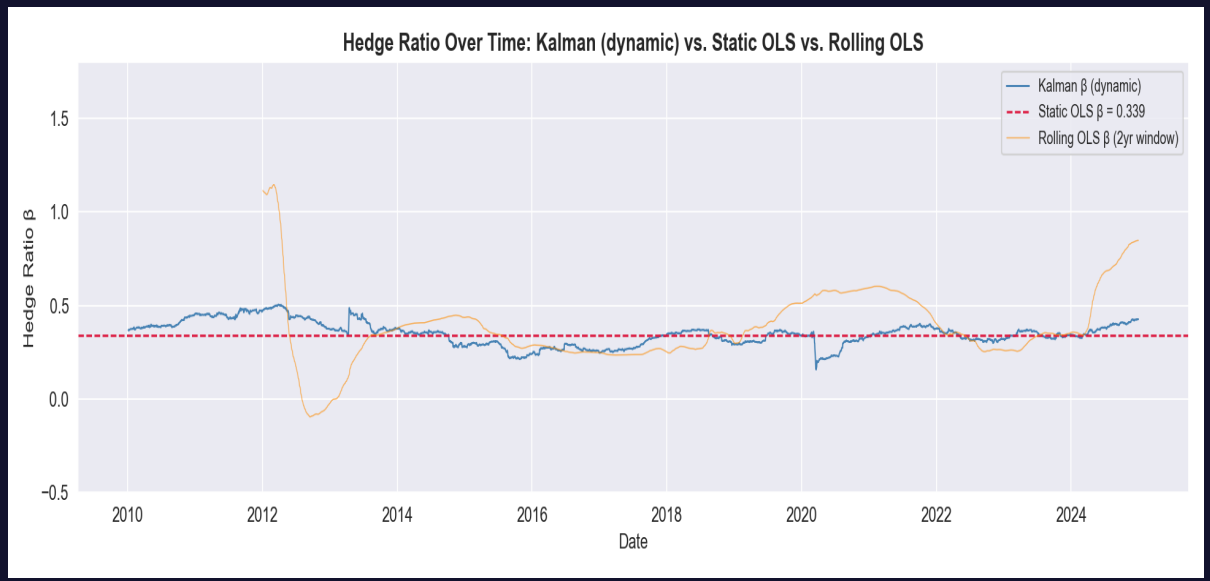

Fig 2: Hedge ratio. Red = static OLS (0.34). Blue = Kalman dynamic (0.16-0.51). Orange = rolling OLS beta (2-year window).

KEY TAKEAWAYS

1 Cointegration is regime-dependent

Active in ~79% of rolling 2-year windows but absent over the full 2010-2024 period. A structural de-rating of gold mining equities after 2013 -- not a data problem -- is why the full-period test fails. The relationship was real and tradeable for extended stretches but macro forces periodically broke it down.

2 Static hedge ratio is inadequate

β drifted from −0.10 to 1.15 across the sample. A single OLS estimate of 0.34 is wrong for most individual periods. The Kalman filter tracks this drift in real time and produces a stationary spread (ADF $p < 0.0001$ vs. 0.87 for the static spread).

3 Regime-aware gating is more coherent

Signals only fire when the rolling cointegration test confirms the relationship is active. 88.7% of days are flat or gated -- the strategy never trades into a non-mean-reverting spread. 110 trades generated over 14 years (59 long, 51 short).

79%	p < 0.0001	23.3%	110
cointegration windows	Kalman spread ADF	ARIMA RMSE gain vs naive	regime-gated trades